

Yu Bin Lee (Yubin Lee)



Korea Institute of Toxicology
141, Gajeong-ro, Yuseong-gu, Daejeon, 34114, Korea
Email: yubin.lee@kitox.re.kr
Tel: +82-42-610-8111

Education

- 2012-2017.02 Ph.D. Bioengineering, Hanyang University, Seoul, Republic of Korea
(Advisor: Prof. Heungsoo Shin Ph.D.)
- 2010-2012 M.S. Bioengineering, Hanyang University, Seoul, Republic of Korea
(Advisor: Prof. Heungsoo Shin Ph.D.)
- 2006-2010 B.S. Bioengineering, Hanyang University, Seoul, Republic of Korea, (Cum Laude)
(Advisor: Prof. Heungsoo Shin Ph.D.)

Professional career

- 2021.05-present Senior Researcher, Korea Institute of Toxicology, Republic of Korea
- 2020.01-2021.04 Principal Scientist, Samsung Electronics, Republic of Korea
- 2019.01-2019.12 Research Associate, University of Illinois at Chicago, USA
(Advisor: Prof. Eben Alsberg Ph.D.)
- 2017.09-2018.12 Research Associate, Case Western Reserve University, USA
(Advisor: Prof. Eben Alsberg Ph.D.)
- 2017.03-2017.08 Post-doctoral researcher, Hanyang University, Seoul, Republic of Korea
(Advisor: Prof. Heungsoo Shin Ph.D.)

Honors and awards

- Outstanding Researcher Award, Korea Institute of Toxicology (2025)
- Korean National Research Council of Science & Technology Young Researcher Award (2024)
- Outstanding presentation award, US-Korea Conference (UKC) 2019, Chicago, USA (2019)
- Outstanding presentation award, Annual Fall and International Symposium of The Korean Society for Biomaterials, Seoul, Republic of Korea (2016)
- Outstanding poster award, Annual Spring Meeting of The Korean Society for Biomaterials, Gwangju, Republic of Korea (2014)
- Brain Korea 21 Scholarship (2013-2015)
- Teaching Assistant Scholarship (2013)
- Hanyang Science & Technology Fellowship, Hanyang University. (2010-2012)
- Undergraduate Research Program (2011)
- Honor Student Award, Hanyang University. (2010)
- National Science & Technology Scholarship, Korea. (2006-2010)

Publications

- 1 J.-K. Kim*, Y.-J. Kim*, S. Ryu., D. Park, K.-S. Moon, **Y.B. Lee***, Dong Ho Woo*, "A 3D Astrocyte Microenvironment Model Enables Rapid Ca²⁺-Resolved Analysis and Therapeutic Modulation of Oxidative Neuroinflammation", *Advanced Healthcare Materials*, e04129, 2026 (**Inside front cover**)
- 2 M. Jo, S.Y. Jeong, M.-I. Kyun, J. Lee, C.H. Choi, **Y.B. Lee***, M. Choi*, J. Rho*, K.-S. Moon*, "Two-Step Algorithmic Selection of Interspecies Sequence Mismatch-Based Housekeeping Genes for Precise Gene Expression Assessment in PBMC-Humanized NSG Mice", *ACS Omega*, 11, 9187, 2026
- 3 A.-R. Lee*, H.Y. Kang*, **Y.B. Lee***, S.Y. Lee, M.-H. Yoo, S.-M. Kim, W.-J. Im, J.-Y. Kim, K.-S. Moon, T.-D. Kim, S.-J. Park, "Safety and efficacy of immune cell therapeutics targeting α -PD-L1 in lung cancer", *Biomedicine & Pharmacotherapy*, 192, 118594, 2025

- 4 M.-I. Kyun*, D. Park*, H. Jung, I. Kim, S.Y. Jeong, J. Kim, M. Jo, J.-I. Kwon, J.K. Kim, D. Kim, S.A. Park, C.S. Ryu, S.K. Kim, D. Park*, **Y.B. Lee***, K.-S. Moon*, “3D Liver Model for Assessing Immune-Mediated Hepatotoxicity from Biopharmaceutical Risks”, *Chemical Engineering Journal*, 523, 1, 2025
- 5 S. Jo*, **Y.B. Lee***, S.-M. Kim, S.Y. Lee, M. Choi, M.-I. Kyun, S.Y. Jeong, S. Lee, J.H. Kim.1, Y. Kim, Y. J. Kim, S. Park, K.-S. Moon, T.-D. Kim*, “Fine-Tuning Signal Strength in CD5 CAR-NK Cells for Targeted T Cell Cancer Therapy”, *Frontiers in Immunology*, 16, 2025
- 6 M.-I. Kyun*, J.-I. Kwon*, H. Jung, J. Kim, M. Jo, **Y.B. Lee***, K.-S. Moon*, “Developing a Human Immune Endothelial Co-culture Model for Inflammatory Vasculitis and Drug Screening”, *ACS Omega*, 10, 47544-47558, 2025
- 8 **Y.B. Lee***, M.-I. Kyun*, Y.J. Lee, H.-E. Shim, K.M. Huh, S.-W. Kang, “Cyclodextrins as multifunctional tools for advanced biomaterials in tissue repair and regeneration”, *Bioactive Materials*, 49, 627, 2025
- 9 S.J. Lee, O. Jeon, **Y.B. Lee**, D.S. Alt, A. Ding, R. Tang, E. Alsberg, “In situ cell condensation-based cartilage tissue engineering via immediately implantable high-density stem cell core and rapidly degradable shell microgels”, *Advanced Composites and Hybrid Materials*, 8, 330, 2025
- 10 H. Jung*, M.-I. Kyun*, J. Kwon, J. Kim, J.K. Kim, D. Park*, **Y.B. Lee***, K.-S. Moon*, “Amplified response of drug-induced liver fibrosis via immune cell co-culture in a 3D in vitro hepatic fibrosis model”, *Biomaterials Science*, 12, 6351, 2024
- 11 S.Y. Jeong, D. Park, T. Park, J.-S. Han, J. Lee, C.H. Choi, M. Jo, **Y.B. Lee**, M.-I. Kyun, M. Choi, D. Park, K.-S. Moon, “Interspecies transcriptome profiles of human T cell activation and liver inflammation in a xenogeneic graft-versus-host disease model”, *Heliyon*, 10, e40559, 2024
- 12 P.C. Wong, K.H. Chen, W.R. Wang, C.Y. Chen, Y.T. Wang, **Y.B. Lee**, J.L. Wu, “Injectable ChitHCl-DDA tissue adhesive with high adhesive strength and biocompatibility for torn meniscus repair and regeneration”, *International Journal of Biological Macromolecules*, 270, 132409, 2024
- 13 I. Kim*, M.-I. Kyun*, H. Jung, J.-I. Kwon, J. Kim, J.-K. Kim, **Y.B. Lee***, Y.-I. Kwon*, K.-S. Moon*, “In Vitro Non-Alcoholic Fatty Liver Disease Model Elucidating the Effect of Immune Environment on Disease Progression and Alleviation”, *ACS Omega*, 9, 20594-25105, 2024
- 14 M. Kyun, T. Park, H. Jung, I. Kim, J.-I. Kwon, S.Y. Jeong, M. Choi, D. Park, **Y.B. Lee***, K.-S. Moon*, “Development of an In Vitro Model for Inflammation Mediated Renal Toxicity Using 3D Renal Tubules and Co-Cultured Human Immune Cells”, *Tissue Engineering and Regenerative Medicine*, 20, 1173-1190, 2023
- 15 **Y.B. Lee***, S. Kim*, E.M. Kim, H. Byun, H. Shin, “Harvest of cell-only muscle fibers using thermally expandable hydrogels with adhesive patterns”, *Tissue Engineering Part C: Methods*, 29, 447-458, 2023
- 16 S.J. Gwak*, **Y.B. Lee***, E.J. Lee, K.H. Park, S.W. Kang, K.M. Huh, “The use of acetylation to improve the performance of hyaluronic acid-based dermal filler”, *Korean J. Chem. Eng.*, 40, 1963-1969, 2023
- 17 T.T. Truong*, **Y.B. Lee***, K.H. Park, H.-E. Shim, J.J. Song, H.-S. Kim, J.H. Hwang, S.-W. Kang, K.M. Huh, “Cryopreservable three-dimensional spheroid culture for ready-to-use systems”, *Korean J. Chem. Eng.*, 40, 390-397, 2023
- 18 W.B. Ahn*, **Y.B. Lee***, Y.-h. Ji, K.-S. Moon, H.-S. Jang, S.-W. Kang, “Decellularized human adipose tissue as an alternative graft material for bone regeneration”, *Tissue Engineering and Regenerative Medicine*, 19, 1089-1098, 2022
- 19 D.-E. Kim*, **Y.B. Lee***, J. J. Song, J.-S. Han, K.-S. Moon, K.M. Huh, S.-W. Kang, “Application of HGC as a non-cell adhesive polymer in three-dimensional cell culture”, *ACS Omega*, 7, 18471, 2022
- 20 O. Jeon, **Y.B. Lee**, S.J. Lee, N. Guliyeva, J. Lee, E. Alsberg, “Stem cell-laden hydrogel bioink for generation of high resolution and fidelity engineered tissues with complex geometries”, *Bioactive Materials*, 15, 185-193 2021
- 21 J. Nulty, F. Freeman, D. Browe, R. Burdis, D. Ahern, P. Pitacco, **Y.B. Lee**, E. Alsberg, D. Kelly, “3D bioprinting of prevascularised implants for the repair of critically-sized bone defects”, *Acta Biomaterialia*, 126, 154-169, 2021
- 22 A. Ding, O. Jeon, R. Tang, **Y.B. Lee**, S.J. Lee, E. Alsberg, “Cell-Laden Multiple-Step and Reversible 4D Hydrogel Actuators to Mimic Dynamic Tissue Morphogenesis”, *Advanced Science*, 2004616, 2021
- 23 **Y.B. Lee**, O. Jeon, S.J. Lee, A. Ding, D. Wells, E. Alsberg, “Induction of Four-Dimensional Spatiotemporal Geometric Transformations in High Cell Density Tissues via Shape-Changing Hydrogels”, *Advanced Functional Materials*, 2010104, 2021 (Back cover, Study of the year)

- 24 B.P. Hung, T. Gonzalez-Fernandez, J.B. Lin, T. Campbell, **Y.B. Lee**, A. Panitch, E. Alsberg, J. K. Leach, "Multi-peptide presentation and hydrogel mechanics jointly enhance therapeutic duo-potential of entrapped stromal cells", *Biomaterials*, 245, 119973, 2020
- 25 J. Lee, O. Jeon, M. Kong, A.A. Abdeen, J.Y. Shin, H.N. Lee, **Y.B. Lee**, W. Sun, P. Bandaru, D.S. Alt, K. Lee, H.-J. Kim, S. J. Lee, S. Chaterji, S. R. Shin, E. Alsberg, A. Khademhosseini, "Combinatorial screening of biochemical and physical signals for phenotypic regulation of stem cell-based cartilage tissue engineering", *Science Advances*, 6, eaaz5913, 2020
- 26 E.M. Kim, **Y.B. Lee**, S. Kim, J. Park, J. Lee, S.W. Kim, H. Park, H. Shin, "Fabrication of core-shell spheroids as building blocks for engineering 3D complex vascularized tissue", *Acta biomaterialia*, 100, 158-172, 2019
- 27 O. Jeon*, **Y.B. Lee***, H. Jeong, S.J. Lee, E. Alsberg, "Individual cell-only bioink and photocurable supporting medium for 3D printing and generation of engineered tissues with complex geometries", *Materials Horizons*, 2019 (Front cover)
- 28 H. Byun, **Y.B. Lee**, E.M. Kim, H. Shin, "Fabrication of size-controllable human mesenchymal stromal cell spheroids from micro-scaled cell sheets", *Biofabrication*, 11, 035025, 2019
- 29 O. Jeon, **Y.B. Lee**, T.J. Hinton, A.W. Feinberg, E. Alsberg, "Cryopreserved cell-laden alginate microgel bioink for 3D bioprinting of living tissues" *Materials Today Chemistry*, 12, 61-70, 2019
- 30 S. Herberg, A. McDermott, P. Dang, D. Alt, R. Tang, J. Dawahare, D. Varghai, J.-Y. Shin, A. Mcmillan, A. Dikina, F. He, **Y.B. Lee**, K. Umemori, P.-C. Wong, H. Park, J. Boerckel, E. Alsberg, "Combinatorial morphogenetic and mechanical cues to mimic bone development for defect repair", *Science advances*, 5, eaax2476, 2019
- 31 E.M. Kim, **Y.B. Lee**, H. Byun, H.-k. Chang, J. Park, H. Shin, "Fabrication of Spheroids with Uniform Size by Self-Assembly of a Micro-Scaled Cell Sheet (μ CS): The Effect of Cell Contraction on Spheroid Formation", *ACS Applied Materials and Interfaces*, 11, 2802-2813, 2019
- 32 **Y.B. Lee***, E.M. Kim*, H. Byun*, H. Shin, "Engineering spheroids potentiating cell-cell and cell-ECM interactions by self-assembly of stem cell microlayer", *Biomaterials*, 165, 105-120, 2018
- 33 **Y.B. Lee**, J.-Y. Lee, T. Ahmad, H. Byun, M. Matsusaki, H. Shin, "One-step delivery of a functional multi-layered cell sheet using a thermally expandable hydrogel with controlled presentation of cell adhesive proteins", *Biofabrication*, 10, 025001, 2018
- 34 I. Jun, T. Ahmad, S. Bak, J.-Y. Lee, E.M. Kim, J. Lee, **Y.B. Lee**, H. Jeong, H. Jeon, H. Shin, "Spatially Assembled Bilayer Cell Sheets of Stem Cells and Endothelial Cells Using Thermosensitive Hydrogels for Therapeutic Angiogenesis" *Advanced Healthcare Materials*, 6, 1601340, 2017
- 35 S.K.M. Perikamana, Y.M. Shin, J.K. Lee, **Y.B. Lee**, Y. Heo, T. Ahmad, S.Y. Park, J. Shin, K.M. Park, H.S. Jung, S.-W. Cho, H. Shin, "Graded functionalization of biomaterial surfaces using mussel-inspired adhesive coating of polydopamine", *Colloids and Surfaces B: Biointerfaces*, 159, 546-556, 2017
- 36 **Y.B. Lee***, S.-J. Kim*, E.M. Kim, H. Shin, "Microcontact printing of polydopamine on thermally expandable hydrogels for controlled cell adhesion and delivery of geometrically defined microtissues.", *Acta Biomaterialia*, 61, 75-87, 2017
- 37 **Y.B. Lee***, Y.M. Shin*, E.M. Kim, J.-Y. Lee, J. Lim, S.K. Kwon, H. Shin, "Mussel adhesive protein inspired coating on temperature-responsive hydrogels for cell sheet engineering", *Journal of Materials Chemistry B*, 4, 6012-6022, 2016
- 38 S. Bak*, T. Ahmad*, **Y.B. Lee**, J.-Y. Lee, E.M. Kim, and H. Shin, "Delivery of cell patch co-cultured with endothelial cells and smooth muscle cells using thermoresponsive hydrogels for enhanced angiogenesis", *Tissue Engineering Part A*, 22(1-2), 182-193, 2016
- 39 I. Jun, Y.-W. Chung, Y.-H. Heo, H.-S. Han, J. Park, K. Kim, **Y.B. Lee**, H. Lee, Y.-C. Kim, H.-K. Seok, H. Shin, H. Jeon, "Creating hierarchical topography on fibrous platforms using femtosecond laser ablation for directing myoblast behaviors", *ACS Applied Materials & Interfaces*, 10;8(5), 3407-3417, 2016
- 40 **Y.B. Lee***, Y.M. Shin*, E.M. Kim, J. Lim, J.-Y. Lee, and H. Shin, "Facile cell sheet harvest and translocation mediated by a thermally expandable hydrogel with controlled cell adhesion", *Advanced Healthcare Materials*, 5, 2320-2324, 2016 (Front cover)
- 41 J. Lim, I.D. Jun, **Y.B. Lee**, E.M. Kim, D. Shin, H. Jeon, H. Park, and H. Shin, "Fabrication of cell sheets with anisotropically aligned myotubes using thermally expandable micropatterned hydrogels", *Macromolecular Research*, 24, 562-572, 2016 (Front cover)
- 42 **Y.B. Lee***, J.-Y. Lee*, T. Ahmad, S. Bak, J. Lee, H. Kim, T.H. Park, N.S. Hwang, Y.M. Lim, and H. Shin, "Construction of 3-D Cellular Multi-Layers with Extracellular Matrix Assembly Using Magnetic Nanoparticles", *Journal of Biomedical Nanotechnology*, 2016

- 43 I. Jun, **Y.B. Lee**, Y.S. Choi, A.J. Engler, H. Park, H. Shin, "Transfer stamping of human mesenchymal stem cell patches using thermally expandable hydrogels with tunable cell-adhesive properties", *Biomaterials*, 54, 44-54, 2015
- 44 S. MP, J. Lee, **Y.B. Lee**, Y.M. Shin, E. Lee, A.G. Mikos, and H. Shin, "Materials from mussel-inspired chemistry for cell and tissue engineering applications", *Biomacromolecules*, 14, 16(9), 2541-2555, 2015
- 45 Y. Heo, Y.M. Shin, **Y.B. Lee**, Y.M. Lim, and H. Shin, "Effect of immobilized collagen type IV on biological properties of endothelial cells for the enhanced endothelialization of synthetic vascular graft materials", *Colloids and Surfaces B: Biointerfaces*, 134, 196-203, 2015
- 46 **Y.B. Lee**, I.D. Jun, S.W. Bak, Y.M. Shin, Y.-M. Lim, H. Park, and H. Shin, "Reconstruction of vascular structure with multicellular components using cell transfer printing methods", *Advanced Healthcare Materials*, 3, 1465-1474, 2014
- 47 S.J. Kim, I.D. Jun, D.W. Kim, **Y.B. Lee**, Y.J. Lee, J.H. Lee, K.D. Park, H. Park, and H. Shin, "Rapid transfer of endothelial cell sheet using a thermosensitive hydrogel and its effect on therapeutic angiogenesis", *Biomacromolecules*, 14, 4309 - 4319, 2013
- 48 **Y.B. Lee**, Y.M. Shin, J.H. Lee, I.D. Jun, J.K. Kang, J.-C. Park, and H. Shin, "Polydopamine-mediated immobilization of multiple bioactive molecules for the development of functional vascular graft materials", *Biomaterials*, 33, 8343-8352, 2012
- 49 K. Yang, J.S. Lee, J. Kim, **Y.B. Lee**, H. Shin, Um SH, J.B. Kim, K.I. Park, H. Lee, and S.-W. Cho, "Polydopamine-mediated surface modification of scaffold materials for human neural stem cell engineering", *Biomaterials*, 33, 6952-6964, 2012
- 50 Y.M. Shin, **Y.B. Lee**, S.J. Kim, J.K. Kang, J.C. Park, W. Jang. and H. Shin, "Mussel-inspired immobilization of vascular endothelial cell growth factor (VEGF) for enhanced endothelialization of vascular grafts", *Biomacromolecules*, 13, 2020-2028, 2012
- 51 S.H. Bhang, S.I. Jeong, T-J Lee, I.D. Jun, **Y.B. Lee**, B.-S. Kim, and H. Shin, "Electroactive polyaniline/poly(L-lactide-co- ϵ -caprolactone) electrospun fibers for control of neural cell function", *Macromolecular Bioscience*, 12, 402-411, 2012
- 52 Y.M. Shin, **Y.B. Lee**, H. Shin, "Time-dependent mussel-inspired functionalization of poly(l-lactide-co- ϵ -caprolactone) substrates for tunable cell behaviors", *Colloids and Surfaces B: Biointerfaces*, 87, 79-87, 2011
- 53 J.H. Lee, **Y.B. Lee**, N.G. Rim, S.Y. Cho, Y.M. Lim, and H. Shin, "Development and characterization of nanofibrous poly(lactic-co-glycolic acid)/biphasic calcium phosphate composite scaffolds for enhanced osteogenic differentiation", *Macromolecular Research*, 19, 172-179, 2011
- 54 N.G. Rim, **Y.B. Lee**, S.J. Kim and H. Shin, "Current status and prospect of biomaterials tissue substitutes in regenerative medicine", *KIC (The Korean Society of Industrial and Engineering Chemistry) News*, 13(6), 2-17, 2010

References

Heungsoo Shin, Ph.D.
 Professor
 Department of Bioengineering
 Hanyang University, Seoul, Republic of Korea
 hshin@hanyang.ac.kr

Eben Alsberg, Ph.D.
 Professor
 Department of Bioengineering
 University of Illinois at Chicago, USA
 ealsberg@uic.edu

Kang Moo Huh, Ph.D.
 Professor of Department of Polymer Science and Engineering
 Chungnam National University
 Daejeon, Korea
 E-mail: khuh@cnu.ac.kr